

Accelerating EvidenceGrounded FAQ Assistants with a Distributed Agentic GraphRAG Framework

Course: Parallel & Distributed Computing (PDC) + Natural Language Processing
(NLP) — Group Project (MidTerm Evaluation)

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Primary Context:

Group Project Proposal (attached): *Accelerating EvidenceGrounded FAQ Assistants: A Distributed Agentic RAG Framework with Parallelized GraphRetrieval and MultiLevel Citations* (Feb 28, 2025)

Executive Summary

Advanced Retrieval Augmented Generation (RAG) improves factual grounding by retrieving supporting evidence before generating answers, but realtime deployment becomes difficult when pipelines add (i) agentic selfcorrection loops (retrieve → critique → refine) and (ii) graphbased retrieval for multihop evidence. Foundational RAG work explicitly motivates provenance/grounding and shows how retrieval-conditioned generation outperforms parametric-only baselines in knowledgeintensive tasks (Lewis et al., 2020). However, iterative agentic workflows including criticdriven corrections increase total retrieval and inference calls, raising latency (Dong et al., 2025).

This midterm partial research report proposes a Distributed Agentic GraphRAG system aimed at evidence-grounded FAQ answering (medical-focused in the proposal), with a primary PDC goal of reducing “stopandwait” bottlenecks through parallel and distributed execution, and a primary NLP goal of improving citation reliability and faithfulness through graph-structured evidence and critique-guided validation. The design draws on GraphRAG formulations, which organize corpora into graph structures for indexing, retrieval, and generation (Peng et al., 2024/2025), and on medical GraphRAG methods such as Triple Graph Construction and URetrieval for evidence-based medical answers (Wu et al., 2025).

Evaluation will measure both systems performance (time to first token, p50/p95 latency, throughput, speedup/efficiency) and NLP quality (retrieval recall, answer correctness on benchmark tasks, and citation metrics inspired by ALCE and RAGAS). The expected contribution is a reproducible blueprint for making agentic GraphRAG pipelines faster without degrading evidence quality.

Introduction

Background and context

Retrieval-Augmented Generation combines a retriever (to fetch relevant external evidence) and a generator (to synthesize an answer conditioned on that evidence). In the original RAG formulation, retrieval supplies nonparametric memory (e.g., Wikipedia passages) that improves factuality, specificity, and performance on knowledge-intensive tasks, while also providing a pathway toward provenance (Lewis et al., 2020). Dense Passage Retrieval (DPR) shows that dense dual encoder retrievers can outperform sparse baselines (e.g., BM25) and substantially improve tool retrieval accuracy in open domain QA settings (Karpukhin et al., 2020).

More recent pipelines go beyond “flat” passage retrieval by adding graph structure (entities, relations, citations, document links) and agentic control (planning and critique) to support multihop reasoning and reduce hallucinations. GraphRAG surveys formalize a workflow consisting of graphbased indexing, graphguided retrieval, and graph enhanced generation, emphasizing how graph structure helps retrieve *connected evidence* rather than isolated text chunks (Peng et al., 2024/2025).

Problem statement

The attached proposal motivates this project with two tightly coupled issues:

Inference latency problem: Agentic RAG workflows often rely on sequential retrieve–critique–refine cycles. Adding graph retrieval and citation verification increases compute and I/O, making realtime FAQ deployment difficult at scale.

Citation reliability problem: Users especially in highstakes domains like healthcare need answers whose claims can be verified via citations. Citationaware generation is itself computationally expensive, and citation quality requires dedicated evaluation (Gao et al., 2023).

This report addresses the combined question: How can we design an evidencegrounded, citationverifiable FAQ assistant that uses agentic GraphRAG methods while remaining fast enough for interactive use?

Project objectives, scope, and significance

Objectives (derived from proposal goals and midterm deliverable needs): Design and implement a distributed agentic GraphRAG pipeline for evidence grounded FAQ answering. Reduce end-to-end latency by exploiting parallel retrieval, parallel critique, and distributed ingestion/indexing, while reporting

speedup/efficiency using established PDC performance laws (Amdahl, 1967; Gustafson, 1988). Improve evidence grounding and citation reliability by integrating: graphstructured evidence (GraphRAG concepts), medical GraphRAG techniques such as Triple Graph Construction and URetrieval (when medical domain is used), critic-guided refinement for error detection and correction (Dong et al., 2025). Evaluate citation quality using benchmark inspired methodologies (ALCE) and componentlevel evaluation (RAGAS).

Scope: This midterm partial report covers Introduction, Literature Review/Related Work, and a detailed Proposed Methodology (implementation-ready plan with pseudocode, diagrams, and code scaffolds). Fullscale experiments and final results are scoped for the final submission.

Significance: Medical consumer QA and clinical support FAQ systems are a natural application domain because hallucinations and unsupported claims create risk; integrating external medical knowledge and citations is repeatedly highlighted as crucial for reducing hallucinations in healthcareoriented LLM systems (Chen et al., 2025).

Research questions and hypotheses

RQ1 (Latency and scalability): How much can distributed task parallelism reduce time to first token and end to end latency for agentic GraphRAG compared with an equivalent serial pipeline?

H1: End to end latency improves substantially when retrieval workers and critics run concurrently, but measured speedup will be bounded by serial fractions (Amdahl, 1967) and may follow scaled workload behavior (Gustafson, 1988).

RQ2 (Citation reliability): Does adding graph structured evidence and critic guided checking improve citation precision/coverage compared with nongraph or noncritic baselines?

H2: Graphguided retrieval improves evidence connectivity and multihop support (Peng et al., 2024/2025) while critic workflows reduce unsupported or misattributed claims (Dong et al., 2025), yielding improved citation metrics on citation oriented evaluation setups (Gao et al., 2023).

RQ3 (Multisource planning): Does explicit source planning reduce retrieval mismatch and improve multisource utilization?

H3: Modeling retrieval as a source planning problem improves alignment to heterogeneous medical sources (Chen et al., 2025), which should transfer to better evidence selection and citation completeness.

Literature Review and Related Work

This literature review is organized thematically and ends with a gap analysis that motivates the proposed system.

RAG foundations and dense retrieval backbones

RAG's central idea is that retrieval allows models to consult external evidence dynamically, improving factuality and addressing knowledge update/provenance limitations of parametric only language models (Lewis et al., 2020). DPR demonstrates that dense retrieval can be implemented effectively with a dual encoder design: a question encoder and a passage encoder produce embeddings for approximate nearest neighbor search, improving retrieval accuracy and enabling strong end-to-end QA pipelines (Karpukhin et al., 2020).

Critical implication for this project: Even with strong retrievers, *grounding failures* occur when retrieved context is incomplete, irrelevant, or not properly connected to the generated claims. This becomes more critical in longform answers where multiple claims require multiple sources, motivating citationaware evaluation and structured evidence selection (Gao et al., 2023).

Citationaware generation and evaluation

ALCE formalizes an end-to-end task in which systems must retrieve evidence and generate outputs with citations, and proposes automatic metrics along fluency, correctness, and citation quality dimensions (Gao et al., 2023). This is directly relevant because the proposal emphasizes multilevel citations and measurable citation accuracy.

RAGAS complements benchmarked evaluation by proposing referencefree metrics to evaluate RAG components such as faithfulness and context relevance, enabling iterative development even when gold references are limited (Es et al., 2023/2024).

Critical implication: Citation quality is not automatically guaranteed by retrieval; it must be measured and often explicitly optimized. However, the measurement itself can increase compute cost if it relies on additional model calls, making systemslevel efficiency central to practical deployment.

GraphRAG and graphguided retrieval

GraphRAG surveys propose that graph structure improves retrieval by capturing relationships between documents, entities, and concepts, enabling multihop or “networked” evidence retrieval that flat top chunk retrieval often misses (Peng et al., 2024/2025). Related directions such as GRAG emphasize retrieving textual subgraphs and integrating both topology and content into LLMs, underscoring the broader trend toward graphstructured retrieval for connected corpora (Hu et al., 2025).

Limitations relevant to this project: Graphbased indexing and traversal add overhead. Many GraphRAG works prioritize quality and retrieval coverage more than interactive latency. This creates a natural “PDC gap”: graph retrieval must be engineered for parallelism, caching, and efficient partitioning to reach interactive speeds.

Domainspecific GraphRAG in medical QA

The proposal’s target domain emphasizes medical FAQs. MedQA and PubMedQA are common benchmarks for medical QA; MedQA is a largescale multichoice dataset sourced from medical board exams (Jin et al., 2020/2021), while PubMedQA is built from PubMed abstracts with yes/no/maybe labels and includes expertannotated and unlabeled/synthetic subsets (Jin et al., 2019). ASQA targets longform answers to ambiguous questions and motivates evaluation for synthesis across multiple sources (Stelmakh et al., 2022).

MedGraphRAG proposes a medical GraphRAG framework designed for evidence-based medical responses, introducing Triple Graph Construction and URetrieval—which aligns *directly* with the proposal’s methodology language (Wu et al., 2025). The project’s plan to exploit graphstructured evidence is therefore grounded in strong domainspecific precedent.

Agentic workflows and critical guided correction

RAGCritic provides a concrete framework for criticized improvement: it defines a hierarchical error system for RAG and progressively aligns an error critical model that provides fine-grained feedback to enhance RAG performance autonomously (Dong et al., 2025). From a systems viewpoint, however, adding critics can multiply LLM calls, potentially worsening latency the core bottleneck the proposal identifies.

OmniRAG reframes multisource medical retrieval as a source planning problem and introduces MedOmniKB plus an optimization method to improve source planning alignment (Chen et al., 2025). Source planning is conceptually compatible with an agentic planner module, but again introduces compute overhead, motivating parallel execution.

PDC framing for accelerating agentic GraphRAG pipelines

In PDC, performance is routinely analyzed via speedup and efficiency. Amdahl's Law formalizes how serial portions cap maximum speedup for a fixed problem size (Amdahl, 1967), while Gustafson's Law argues that scaled workloads can achieve higher effective speedup as parallel resources increase (Gustafson, 1988).

Modern distributed Python runtimes operationalize this by executing task graphs across worker pools. Dask explicitly models computations as task graphs and provides multiple schedulers with differing tradeoffs (ask documentation on task graphs and scheduling). Ray provides task/actor abstractions and fault tolerance mechanisms such as retries and actor restarts, which are relevant when retrieval workers or index services fail or scale dynamically (Ray fault tolerance documentation).

Gap analysis and motivation for this project

Gap: Many high-quality agentic and GraphRAG systems remain expensive at inference time due to multistep loops and graph traversal overhead.

Consequence: Interactive FAQ use cases suffer, especially when answers require multihop evidence and citations. TTFT and p95 latency become critical userfacing constraints.

Project response: Build an agentic GraphRAG pipeline explicitly engineered as a distributed task graph: parallel retrieval workers, parallel evidence checks, distributed ingestion/indexing, and scalability reporting through speedup/efficiency.

Proposed Methodology

This section describes the target end-to-end system and provides working details sufficient to begin implementation.

System overview

The proposed system is a Distributed Agentic GraphRAG FAQ assistant consisting of five stages:

Ingestion (distributed): Parse and chunk raw documents; extract metadata and optionally entities/definitions.

Indexing (parallel): Build a vector index for dense retrieval and a graph structure linking entities, definitions, and source chunks.

Planning (agentic): A planner decomposes the user query and selects retrieval strategies (multisource planning).

Retrieval (parallel): Multiple retrieval workers run concurrently (dense ANN, graphguided expansion, sparse/keyword retrieval).

Critique and generation (agentic): A critic checks evidence sufficiency and citation support; the generator produces a grounded answer with citations.

Data sources, datasets, and splits

Proposal-aligned default (medical FAQ focus): MedQA for medical examstyle QA (multiplechoice). Use English subset and official splits if available; otherwise define an 80/10/10 train/dev/test split stratified by source exam subset.

PubMedQA for evidence-based reasoning over abstracts with yes/no/maybe labels. Use the expertlabeled subset for evaluation and optionally leverage synthetic/unlabeled subsets for retrieval corpus building (with careful leakage control).

ASQA as a longform QA benchmark to evaluate synthesis + citation support across multiple sources (especially relevant for ALCEstyle citation evaluation).

MedOmniKB (from OmniRAG) as a multigenre repository to test source planning; if unavailable, approximate with a curated set of textbooks/guidelines + paper abstracts.

If any dataset access is constrained (unspecified in proposal), choose one of these practical options:

Option A (coursescala subset): Use 5k-20k questions per dataset and a bounded retrieval corpus to enable many ablations quickly.

Option B (fullscale evaluation): Use complete official splits and measure scaling on a larger corpus; requires more compute and stronger distributed indexing.

Option C (retrieval first medical setting): Focus on PubMedQA + a large abstract corpus; measure groundedness/citations more than MCQ accuracy.

Preprocessing and ingestion

Document parsing and chunking: Normalize text, preserve section boundaries where available, and store provenance metadata (document ID, section, offsets).

Apply semantic chunking (tokenaware) to reduce truncation and improve retrievability.

Entity/definition extraction (optional but recommended for graph construction): Biomedical entity recognition and linking can be implemented using display models and optional concept normalization against UMLS Metathesaurus.

Definitions can be extracted via patterns ("X is defined as...") and/or retrieved from a controlled vocabulary graph.

Distributed ingestion approach: Partition corpus by document ID; run parsing/chunking/entity extraction on worker nodes.

Batch compute embeddings to maximize GPU utilization; cache intermediate artifacts (chunks, entity mentions) for reproducibility.

Indexing and graph construction

Vector index: Build a dense embedding index for chunk retrieval. Faiss provides indexing and similarity search primitives, and explicitly targets largescale embedding search workloads.

Graph structure (proposalaligned): Use a triplestyle graph similar to MedGraphRAG: Entity nodes: normalized medical concepts (optionally UMLS concept identifiers).

Definition nodes: definitional text segments (controlled vocabulary or definitional sentences).

Source nodes: chunklevel evidence units with provenance metadata.

This matches MedGraphRAG's "Triple Graph Construction" premise and supports multilevel citation tracing (claim → chunk → document).

Graph storage: If compute/infra is limited, store graph adjacency lists in a lightweight database; if aiming for scalable deployment, use a vector database with metadata filtering and separate graph storage. Milvus is a distributed vector database with an architecture designed for high-performance search over massive embedding collections.

Agentic workflow and parallel retrieval

Planner agent (source planning + query decomposition): Decompose the question into subquestions and decide which retrieval “channels” to query (dense, graph expansion, sparse).

For multisource settings, incorporate OmniRAG’s framing: generate context-appropriate queries aligned with source attributes.

Parallel retrieval workers: Worker A (dense ANN): DPRstyle embeddings + top ANN search.

Worker B (graphguided): retrieve seed chunks then expand via graph neighbors (entities → definitions → linked sources).

Worker C (sparse/keyword): keyword/BM25 or tagindex retrieval to catch rare terms, acronyms, and exact mentions (optional but often useful in medical terminology).

Evidence merging: Normalize scores per channel, deduplicate by chunk/document, then construct an evidence set that balances relevance, diversity, and citation coverage (ALCE motivation).

Critic module (parallel checking): Implement a critic guided by RAGCritic principles: identify error categories such as missing evidence, irrelevant retrieval, or unsupported claim risk, and provide feedback for the next iteration.

Run critic checks concurrently with retrieval planning for the next round when feasible (speculative planning), reducing the “stopandwait” pattern.

Evaluation Plan and Experimental Design

Baselines and ablations

Baselines: Serial agentic GraphRAG baseline: same planner/critic/graph logic but executed sequentially.

Vanilla RAG baseline: singlepass retrieve then generated (Lewis et al., 2020).

Dense retrieval baseline: DPR retrieval + generator without graph expansion (Karpukhin et al., 2020).

Ablations (to isolate contributions): Remove graph expansion (vectoronly). Remove critic loop (no iterative correction). Remove source planning (singlestrategy queries). Increase workers/nodes and report speedup/efficiency.

Metrics table mapping metrics to goals

Goal	Metric	Definition/notes	Source motivation
Fast responsiveness	TTFT (p50/p95)	Includes queuing + prefill - TTFT is a standard interactive network; sensitive to latency metric prompt length [14]	
Full response time	Endtoend latency (p50/p95)	Time until final token important for nonstreaming uses [49]	Complements TTFT
Scalability	Speedup Efficiency E(N)	S(N) Classical PDC reporting for parallel scaling [9]	Validates distributed design
Retrieval quality	Recall@k, nDCG	MRR Measure whether gold/supporting passage is retrieved early [26]	Retrieval determines grounding
Answer correctness	Taskspecific accuracy/F1	MedQA accuracy PubMedQA accuracy ASQA longform metric approach [50]	Measures utility or benchmarks
Citation quality	Citation precision/recall/F1; ALCE metrics	ALCE proposes citation quality metrics and correlates with humans [8]	Verifiability focus
Faithfulness	RAGAS faithfulness, context relevance	Referencefree evaluator for RAG pipelines [17]	Componentlevel debugging

Experimental setup and hardware options

Because compute availability can be unspecified in course projects, the methodology supports multiple setups:

Single workstation: 1 GPU + multicore CPU; local Ray/Dask; Fails indexing; suitable for fast prototyping. [

Small cluster (recommended for PDC claims): 2–8 nodes; Ray cluster or Dask distributed; optional Milvus deployment to test distributed vector search.

Serving-oriented benchmarking: Measure TTFT and throughput under concurrent requests; definitions and measurement guidance can follow industry benchmarking guidance where TTFT includes queue/prefill/network.

Chart suggestions to embed in the final report

Speedup vs number of workers: xaxis = workers (N), yaxis = speedup $S(N) = T(1)/T(N)$, plotted for TTFT and end-to-end latency; include efficiency $E(N) = S(N)/N$. Interpret results with Amdahl and Gustafson.

Citation F1 vs p95 latency tradeoff: xaxis = p95 end-to-end latency; yaxis = citation F1 (ALCEstyle or projectdefined); points correspond to ablations (no critic, critic only, critic+graph, full system).

Risks and limitations

Serial bottlenecks and diminishing returns: Even with extensive parallelism, serial phases (e.g., final generation) bound speedup per Amdahl's Law, so linear scaling is not guaranteed.

Critic overhead: Critic-guided correction improves reliability but can add extra inference calls; this motivates parallel critique and strict iteration budgets.

Dataset availability and leakage: PubMedQA includes labeled, unlabeled, and synthetic subsets; careless use can cause contamination between retrieval corpus and evaluation, so splits and corpus construction must be documented carefully.

Graph construction cost: Building entity-definition-source graphs requires entity extraction and linking; inaccurate linking may introduce noise, so domain tools (scispaCy, UMLS) and ablation studies are needed.

Ethical considerations

Nonclinical use and user safety: Outputs must be framed as informational, not medical advice. Even evidence-grounded systems can fail if evidence is incomplete or misinterpreted; therefore the system must expose citations and encourage verification (ALCE motivation).

Privacy and licensing: If private medical data is used, strict access controls and anonymization are required; if using public benchmarks, ensure licensing compliance and document the corpus sources (MedGraphRAG's medical framing underscores the sensitivity of private medical data usage).

Avoiding fabricated citations: Citation generation must reference only retrieved evidence chunks; the critic should explicitly check for unsupported claims and missing citations.

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